

Data Augmentation Importance for Classification of Skin Lesions via Deep Learning

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Abstract— Melanoma is a fatal type of cancer which is mostly curable if detected in early stages. Various machine learning algorithms are used for distinguishing benign lesions from malignant such as deep learning. To obtain successful result from deep learning, large and quality training data set is essential. But, existing data sets maybe insufficient for training a deep learning network. Building a powerful classifier from insufficient data, data augmentation methods are useful. In this article the same network trained with augmented skin lesion images and non-augmented skin lesion images for detecting malignant skin lesions. When compared results, it has been seen that the network using augmented data for training has achieved better results than training with non-augmented data.

Keywords—deep learning; data augmentation; convolutional neural networks; skin lesions; melanoma

I. INTRODUCTION

Malignant melanoma is a skin disease and one of the fatal type of cancer. Every year thousands of people suffer from this disease. In the United States, approximately 76,380 new cases and 10,130 deaths detected from melanoma in 2016 [1]. If melanoma diagnosed in early stages, it is mostly a curably disease. However, because of similarities of benign and malignant melanomas, it is difficult to distinguish benign skin lesion from malignant one (Fig. 1). This situation may be difficult for even well trained medical expert. Without computer assistance accurate diagnosis rate of melanoma is between 75% to 85% by medical expert [2]. This diagnosis rates decrease when the medical experts not quite well trained. Also, restricted access to medical expert opinion is one of other problems in diagnosis of melanoma. Because of described reasons, automated image analysis and diagnosis systems are an active research field also have critical role of in terms of effective diagnosis process.

While diagnosing melanoma medical experts use different diagnosis methods such as ABCD rule [3] (asymmetry, border, color, diameter, irregularity and texture). These features are used for identifying melanoma. Another identification method is seven-point check list [4]. Described methods and different machine learning algorithms such as support vector machines (SVM), k-nearest neighbors (kNN), artificial neural networks (ANN) utilized to the classification of skin lesions [5].

There is a review about the classification of skin lesions using hand-crafted features and different machine learning algorithms [6].

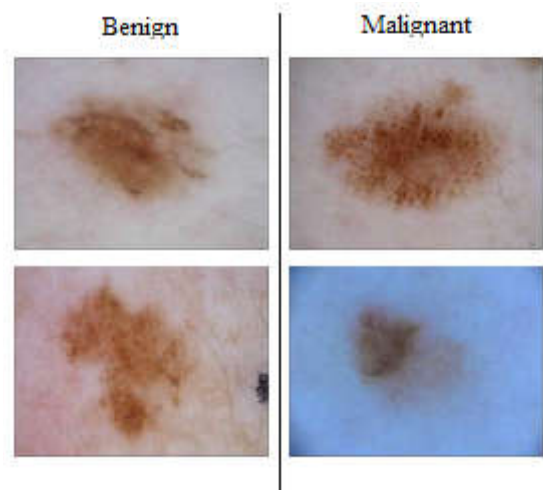


Fig. 1. Samples benign and malignant images from ISIC dataset [7]

Deep learning is a sub field of machine learning and promising successful results in various tasks such as pattern recognition, medical image analysis etc. A review of different applications are given [8] and [9]. Convolutional neural networks (CNN) are very popular among the deep learning approaches which are inspired by human visual cortex. AlexNet won ImageNet competition in 2012 which is a type of CNN [10]. After this success CNNs have been used various image segmentation and classification problems such as in [11] a CNN model is used for segmentation of brain tumors. In other study CNNs are used for the extraction of vessels in fundus images [12]. For Medical images segmentation, different CNNs models are proposed such as U-Net [13].

In classification and segmentation of skin lesions, CNNs are widely used and showed successful results. Codella et. al. used CNN for segmentation and classification of skin lesions for detection of melanoma from dermoscopy images [14]. They acquired %94 accuracy on segmentation and %85 accuracy on classification. In [15] Lopez et. al. used a CNN model for detection of melanoma. They used transfer learning and

acquired % 81 accuracy on dermoscopy images. In [16] Lequan et. al. developed a framework, which is segmenting and classifying melanoma using dermoscopy images, using fully convolutional residual network (FCRN) and deep residual network(DRN). Their accuracy rates are respectively segmentation %94 and classification %85.

Although CNNs have shown remarkable performance, they have some deficits such as overfitting problem and need large amount of training data. There are some methods to avoid overfitting. One of them is training network as many and balanced data as possible. However, creating the data set for the identified problem is a time-consuming and quite laborious task. Existing data sets are often insufficient or unbalanced for the training. Specially, to build a powerful image classifier from little training data, data augmentation methods are very useful.

In this study, the effect of image augmentation methods on CNNs were examined. A CNN model was created and dermoscopy images were used to train the network to achieve the classification of benign and malignant skin lesions. In experiment one network trained with non-augmentation data. In second experiment image augmentation methods was used and two experiment’s accuracy and loss results were compared.

The rest of this paper is organized as in section 2 used CNN architecture and image augmentation methods are explained. The result of the trained networks is discussed in section 3 and conclusions are presented in section 4.

II. METHOTS

In this section, we described proposed CNN architecture, used data augmentation methods and trained data set details.

A. Data Set and Pre Processing

In this study, The International Skin Imaging Collaboration (ISIC) data set was used [7]. The used data set contains 500 benign and 500 malignant skin lesion samples. The data set consists of images with different dimensions such as 1022x767x3, 1504x1129x3, 962x674x3, 2048x1536x3. Before the training all images are resized to 224x224x3 and every pixel divided to 255 for normalized range in [0,1].

B. CNN Architecture

The used CNN architecture is shown in Fig. 2 Every image in data set is resized 224x224x3. As shown in Fig. 2 CNN network is consisting of 11 layers with the following order: conv1, conv2, maxpool1, dropout, conv3, conv4, maxpool2, dropout, dense, dropout, dense. The detailed configuration of network is given in Table I. In convolution layers, Relu is used as activation function and softmax function is used as classifier for fully connected layer output. The network has two outputs as benign or malignant. The same CNN model was used for train with augmented and non-augmented data. Total trainable parameter size is 99,190,306.

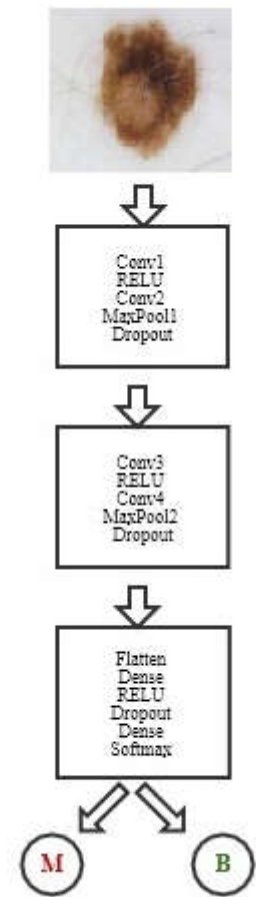


Fig. 2. CNN architecture

TABLE I. CNN architecture details

Layer	Descriptions
Input	224x224x3
Conv1	3x3x32, stride 1, padding=same
Conv2	3x3x32, stride 1, padding=same
MaxPool1	pool_size=2x2
Conv3	3x3x64 stride=1, padding=same
Conv4	3x3x64 stride=1, padding=valid
MaxPool2	pool_size=2x2
Dense1	512
Dense 2	2

C. Data Augmentation

The main idea behind this study is examining performance of CNN network using augmented skin lesion images. In order to increase the accuracy of CNN model, training data amount is very important factor. We used different image augmentation methods such as random transformations, rotations of different angles (20,35), shifting, flipping, zooming etc. Fig. 3 shows

original data and Fig. 4 shows augmented skin lesion examples. We used Keras ImageDataGenerator API for augmenting skin lesion images.



Fig. 3. Original skin lesion

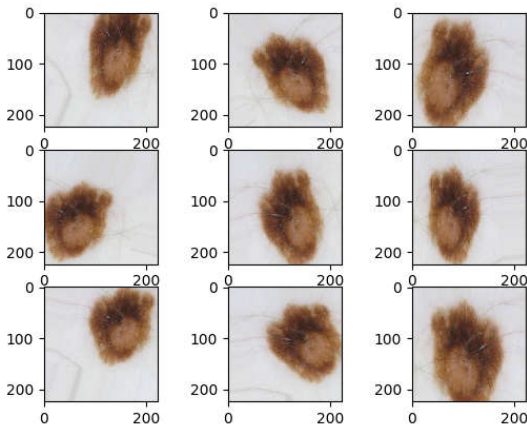


Fig. 4. Augmented skin lesions

III. EXPERIMENTAL RESULTS

In this section, we shared our training details and results. The same CNN architecture was trained twice. In first training procedure, non-augmented data set was used. The used data set contains 1000 skin lesion images (500 benign, 500 melanoma) which 750 images were used for training and 250 images were used for validation and testing. The validation data set was splitted randomly. In the second training procedure augmented data set was used. The augmented data set contains 5000 skin lesion images and 3750 images for training 1250 images validation and test. The proposed CNN model was implemented in Tensorflow and Keras [17] using Python programing language, on a system with two Intel Xenon E5-2640 processor, 64 GB ram and Nvidia Quadro K4200 GPU card. The Stochastic Gradient Descent (SGD) was used as solver[18]. Initial learning rate is 0.001, weight decay set as 1e-4 and momentum set as 0.9. The network trained through 25 epochs and batch size is set as 10. Dropout was used with p=0,5 for avoiding overfitting problem. The binary_crossentropy was used as loss function. Fig. 5 shows trained with augmented data experiment’s accuracy and loss graphics. Fig. 6 shows trained experiment’s accuracy and loss graphics using non-augmented data. Also, Table II gives two experiment’s accuracy and loss rates.

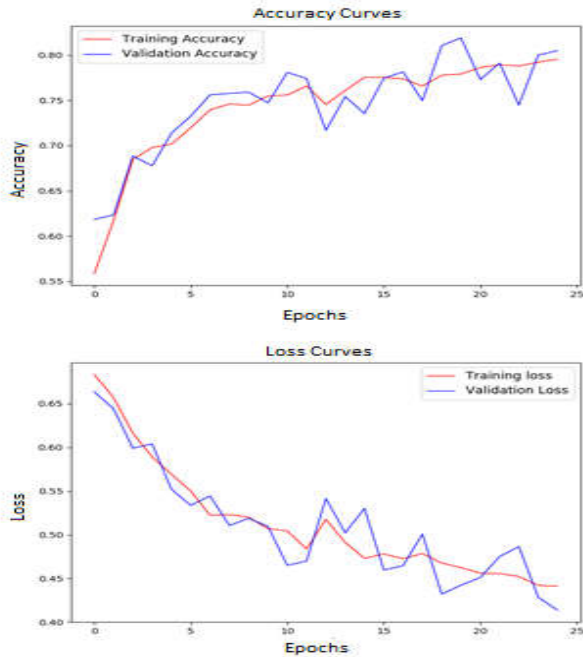


Fig. 5. Training accuracy and loss graphics on augmented data

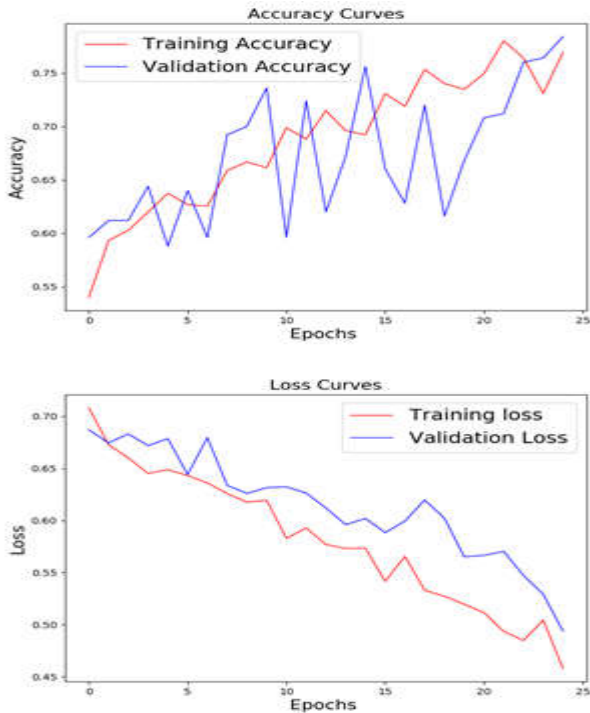


Fig. 6. Training accuracy and loss graphics on non-augmented data

TABLE II. Comparisons of accuracy and loss values

	Accuracy	Loss
Non- Augmented	%78	0,49
Augmented	%81	0,41

IV. CONCLUSION

We proposed the solutions for classification of skin lesions as benign or melanoma under insufficient training data. There is a big demand in deep learning with little training data for obtain high accuracy rates. In this study we trained a CNN architecture with normal data set and augmented data set. Our results show that trained CNN model with augmented data set increased accuracy rates and avoided overfitting (Fig. 5 and Fig. 6). In future work, we will use fine tuning and transfer learning methods in addition to data augmentation to increase the accuracy rate.

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